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United States Patent	12385817
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Matsumoto; Sigeru et al.

Impact test device

Abstract

An impact test device includes a base, a dolly capable of traveling with a test piece placed thereon, and a fall preventing structure configured to prevent the test piece from falling over. The fall preventing structure includes a first section independent of the dolly The first section is provided so as to be movable in a traveling direction of the dolly with respect to the base.

Inventors: **Matsumoto; Sigeru (Tokyo, JP), Miyashita; Hiroshi (Tokyo, JP), Murauchi; Kazuhiro (Tokyo, JP), Haneishi; Kiyoaki (Tokyo, JP)**

Applicant: **KOKUSAI KEISOKUKI KABUSHIKI KAISHA** (Tama, JP)

Family ID: **76574653**

Assignee: **KOKUSAI KEISOKUKI KABUSHIKI KAISHA (Tama, JP)**

Appl. No.: **17/836680**

Filed: **June 09, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20220299412 A1	Sep. 22, 2022

Foreign Application Priority Data

JP	2019-237689	Dec. 27, 2019

Related U.S. Application Data

continuation-in-part parent-doc WO PCT/JP2020/047302 20201217 PENDING child-doc US 17836680

Publication Classification

Int. Cl.: **G01M7/08** (20060101); **G01N3/30** (20060101)

U.S. Cl.:

CPC **G01N3/30** (20130101); G01M7/08 (20130101)

Field of Classification Search

USPC: None

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Primary Examiner: Meier; Stephen D

Assistant Examiner: Parco, Jr.; Ruben C

Attorney, Agent or Firm: Oliff PLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This is a Continuation-in-Part of International Application No. PCT/JP2020/047302 filed on Dec. 17, 2020, which claims priority from Japanese Patent Application No. 2019-237689 filed on Dec. 27, 2019. The entire disclosures of the prior applications are incorporated herein by reference.

BACKGROUND
Technical Field
(1) The present disclosure relates to an impact test device.
Related Art
(2) An impact test device is used to evaluate strengths of products and the adequacies of packaging designs. A conventional impact test device is provided with a support plate that supports a test article in order to prevent the test article from falling over during the test. The support plate is installed to a traveling unit (an impact table) on which the test article is placed to be perpendicular to a traveling direction of the traveling unit.
SUMMARY
(3) In the conventional impact test device described above, since the support plate is installed to the impact table, a weight of the impact table increases by a weight of the support plate. Furthermore, since the support plate having a large surface area is disposed perpendicular to the traveling direction of the impact table, a large wind pressure acts on the support plate when the impact table is traveling at a high speed. The increase in the weight of the impact table due to the installation of the support plate and the increase in the wind pressure the impact table receives during traveling (in other words, an increase in air resistance) cause power required to drive the impact table to increase, resulting in an increase in power consumption. In addition, due to the increase in power required to drive the impact table, a larger motor (i.e., a larger motor) may be required to drive the impact table and may result in problems of an increase in initial cost and an increase in size of the test device.
(4) At least one aspect of the present disclosure is advantageous to reduce the increase in the power required to drive the impact table due to the introduction of a structure for preventing the test article from falling over.
(5) According to aspects of the present disclosure, there is provided an impact test device including a dolly capable of traveling with a test piece placed thereon, and a fall preventing structure configured to prevent the test piece from falling over. The fall preventing structure includes a first section independent of the dolly. The first section is provided so as to be movable in a traveling direction of the dolly.
(6) According to aspects of the present disclosure, there is further provided an impact test device including a dolly capable of traveling with a test piece placed thereon, and a fall preventing structure configured to prevent the test piece from falling over. The fall preventing structure includes second section installed on the dolly. The second section includes a plurality of columnar support erected on the dolly, and a linear member stretched over the plurality of columnar support.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a plan view of an impact test device according to a first embodiment of the present disclosure.

(2) FIG. **2** is a side view of the impact test device according to the first embodiment of the present disclosure.

(3) FIG. **3** is a front view of the impact test device according to the first embodiment of the present disclosure.

(4) FIG. **4** is an enlarged plan view around an impact generating device of the first embodiment.

(5) FIG. **5** is an enlarged side view around the impact generating device of the first embodiment.

(6) FIGS. **6**(*a*) and **6**(*b*) are block diagrams showing schematic configurations of a control system of an impact test device according to illustrative embodiments of the present disclosure.

(7) FIG. **7** is a plan view of an impact test device according to a second embodiment of the present disclosure.

(8) FIG. **8** is a diagram showing a schematic configuration of a modified example of a dolly driver.

(9) FIG. **9** is a diagram showing a schematic configuration of a modified example of a drive module.

DETAILED DESCRIPTION

(10) It is noted that various connections are set forth between elements in the following description. It is noted that these connections in general and, unless specified otherwise, may be direct or indirect and that this specification is not intended to be limiting in this respect.

(11) Hereinafter, illustrative embodiments of the present disclosure will be described with reference to the drawings. In the following description, the same or corresponding numerals are assigned to the same or corresponding components, and redundant descriptions are herein omitted. In each drawing, in a case where a plurality of item whose numerals are in common are shown, the numeral is not necessarily assigned to all of the plurality of items, and assignment of the numeral to some of the plurality of item is appropriately omitted. Furthermore, in each drawing, some components are omitted or shown in cross sections for convenience of explanation.

First Embodiment

(12) FIGS. **1-3** are a plan view, a side view and a front view, in this order, of an impact test device **1** according to a first embodiment of the present disclosure, and FIG. **4** and FIG. **5** are an enlarged plan view and an enlarged side view, in this order, around a later-described impact generating device **40** of the impact test device **1**.

(13) In the following description, a direction from left to right in FIG. **1** is defined as an X-axis direction, a direction from bottom to top in FIG. **1** is defined as a Y-axis direction, and a direction perpendicular to the paper from the back side to the front side is defined as a Z-axis direction. The X-axis direction and the Y-axis direction are horizontal directions orthogonal to each other, and the Z-axis direction is a vertical direction. In addition, the X-axis positive direction is referred to as front, the X-axis negative direction is referred to as back, the Y-axis positive direction is referred to as left, and the Y-axis negative direction is referred to as right.

(14) The impact test device **1** according to the first embodiment of the present disclosure, which will be described below, is configured to be able to perform two types of test methods: a conventional test method (hereinafter referred to as a “collision type test”) in which an impact is applied to a test piece W by causing an impact table on which the test piece W is placed to collide with an impact waveform generating device, and a new test method (hereinafter referred to as a “non-collision type test”) in which an impact is applied to the test piece W by transmitting, to the impact table, an impact generated by driving a motor based on an impact waveform.

(15) The impact test device **1** includes a dolly **20** (an impact table) on which a test piece W is

placed, a dolly driver **30** that drives the dolly **20**, a track unit **10** (a dolly track) that supports the dolly **20** and a later-described carriage **32** of the dolly driver **30** so as to be travelable in the X-axis direction, an impact generating device **40** (an impact waveform generating device) that generates an impact to be applied to the test piece W by a collision with the dolly **20** in the collision type test, and a fall preventing structure that prevents the test piece W from falling over.
(16) The track unit **10** includes a roadbed **11** installed on the base **2** and extending in the X-axis direction, and a guideway-type circulation linear bearing (hereinafter referred to as a "linear guide") **12** installed on the roadbed **11**. The track unit **10** of the present embodiment includes a plurality of set (e.g., three sets) of the roadbed **11** and linear guide **12** arranged for example at equal intervals in the left-right direction (i.e., in the Y-axis direction). It should be noted that a plurality of roadbed **11** may be integrally connected. For example, the track unit **10** may have a configuration in which a plurality of linear guide **12** are installed on a single roadbed **11**.
(17) Each linear guide **12** includes a rail **121** laid on an upper surface of the roadbed **11** (i.e., fixed to the base **2**), and a plurality of (e.g., seven) runner **122** (FIG. **3**) capable of traveling on the rail **121** via a plurality of rolling element. The rolling elements are held in the runners **122** and in a circulating path formed between the rail **121** and the runners **122**. Some of the plurality of runner **122** (e.g., five runners **122**) of each linear guide **12** are arranged for example at equal intervals in the X-axis direction and attached to a lower surface of the dolly **20**. The remaining runners **122** (e.g., two runners **122**) of each linear guide **12** are arranged for example at equal intervals in the X-axis direction and attached to a lower surface of the later-described carriage **32** of the dolly driver **30**. The linear guide **12** guides movements of the dolly **20** and the carriage **32** in the X-axis direction. The carriage **32**, the dolly **20**, and the impact generating device **40** are arranged in this order in the X-axis direction.
(18) The dolly driver **30** includes a carriage **32** that is capable of traveling on the track unit **10**, and a carriage drive section that drives the carriage **32**. The carriage drive section includes a drive module **34** that generates power for driving the carriage **32**, and a belt mechanism **35** that transmits the power generated by the drive module **34** to the carriage **32**.
(19) The dolly driver **30** of the present embodiment includes four drive modules **34** (**34**FL, **34**BL, **34**FR, and **34**BR) disposed near four corners of the base **2**, respectively, and two sets of belt mechanisms **35** (**35**L and **35**R) on the left and on the right, respectively. The belt mechanism **35**L on the left is driven by a pair of drive modules **34**FL and **34**BL on the left, and the belt mechanism **35**R on the right is driven by a pair of drive modules **34**FR and **34**BR on the right.
(20) As shown in FIG. **3**, the drive module **34** includes a frame **341**, a servomotor **342**, a belt mechanism **343**, a shaft **344**, and a plurality of bearing **345**. The servomotor **342** is attached to an upper portion of the frame **341** with a shaft **342***b* thereof oriented in the Y-axis direction. The shaft **344** is disposed below and parallel to the shaft **342***b* of the servomotor **342**, and is rotatably supported by the plurality of bearing **345** attached to the frame **341**.
(21) The belt mechanism **343** includes a drive pulley **343***a* coupled to the shaft **342***b* of the servomotor **342**, a driven pulley **343***c* coupled to the shaft **344**, and a toothed belt **343***b* wound around the drive pulley **343***a* and the driven pulley **343***c*. Each of the drive pulley **343***a* and the driven pulley **343***c* is a toothed pulley having teeth adapted to the toothed belt **343***b*. In the present embodiment, since the number of teeth (i.e., a pitch diameter) of the driven pulley **343***c* is larger than that of the drive pulley **343***a*, the belt mechanism **343** functions as a speed reducer and thus transmits torque output from the servomotor **342** to the shaft **344** while amplifying the torque.
(22) The belt mechanism **35** includes a pair of drive pulleys **351**, a toothed belt **352** wound around the pair of drive pulleys **351**, and a belt clamp **353** (winding intermediate node fixing tool) for fixing the toothed belt **352** to the carriage **32**. The drive pulleys **351** are coupled to the shafts **344** of the drive modules **34**, respectively.
(23) The toothed belt **352** and the toothed belt **343***b* have cores of steel wires. It should be noted that toothed belts that have cores formed of so-called super fibers such as carbon fibers, aramid

fibers, and ultra-high-molecular-weight polyethylene fibers may be used as the toothed belt **352** and the toothed belt **343***b*. By using a lightweight and high-strength core such as a carbon core, the carriage **32** and the dolly **20** can be driven at a large acceleration even when the servomotor **342** having a relatively low output is used, and thus the impact test device **1** can be downsized. Furthermore, in the case of using the servomotor **342** having the same output, by using the lightweight toothed belts **352** and **343***b* having cores formed of the so-called super fiber, it is possible to apply an impact of a larger acceleration to the test piece W.

(24) The pair of drive pulleys **351** of the belt mechanism **35**L on the left are coupled to the shafts **344** of the pair of drive modules **34**FL and **34**BL on the left, respectively, and the pair of drive pulleys **351** of the belt mechanism **35**R on the right are coupled to the shafts **344** of the pair of drive modules **34**FR and **34**BR, on the right, respectively.

(25) The carriage **32** is releasably coupled to a rear end portion of the dolly **20** by coupling parts **321** (e.g., bolts, electromagnets, or the like). Specifically, the carriage **32** is integrally coupled to the dolly **20** when performing the non-collision type test, and is separated from the dolly **20** when performing the collision type test.

(26) The dolly **20** includes a table **21** and a plastic programmer (hereinafter referred to as a "pad") **22** which is a first impact adjuster attached to a front surface of the table **21**. The dolly **20** of the present embodiment includes four pads **22** arranged at equal intervals in the Y-axis direction.

(27) As shown in FIG. **4**, the impact generating device **40** includes a movable block **41**, a track unit **42** (block tracks) that support the movable block **41** so as to be movable in the X-axis direction, pads **43** attached to a back surface (a surface facing the dolly **20**) of the movable block **41**, and two pairs of shock absorbers **44** and **45** disposed on left and right sides of the movable block **41**. The shock absorbers **44** and **45** are, for example, hydraulic shock absorbers.

(28) The track unit **42** includes a roadbed **421** installed on the base **2** and extending in the X-axis direction, and a linear guide **422** installed on the roadbed **421**. The track unit **42** of the present embodiment includes a plurality of set (e.g., three sets) of the roadbed **421** and linear guide **422** arranged at equal intervals in the left-right direction. It should be noted that the plurality of roadbed **421** may be integrally coupled. For example, the track unit **42** may have a configuration in which a plurality of linear guide **422** are installed on a single roadbed **421**.

(29) Each linear guide **422** includes a rail **422***a* laid on an upper surface of the roadbed **421**, and a plurality of (e.g., four) runner **422***b* capable of traveling on the rail **422***a*. The plurality of runner **422***b* are, for example, arranged at equal intervals in the X-axis direction, and are attached to a lower surface of the movable block **41**. The linear guide **422** guides movement of the movable block **41** in the X-axis direction.

(30) As shown in FIGS. **1** and **3**, a pair of linear encoders **744** are provided on left and right sides of the dolly **20** at a central portion of a movable range of the dolly **20**. Main bodies of the linear encoders **744** are attached to the roadbed **11**. Scales **744***b* of the linear encoders **744** are attached to left and right sides of the dolly **20**. A position and a speed of the dolly **20** are detected by the linear encoders **744**.

(31) The impact generating device **40** includes as many pads **43** as the pads **22** of the dolly **20**, the pads **43** being a second impact adjuster. The pads **43** are paired with the pads **22** and attached to a back surface of the movable block **41** at positions facing the corresponding pads **22**.

(32) As shown in FIG. **4**, a pair of push plates **411** having two flat surfaces (operating surfaces **411***a* and **411***b*) perpendicular to the X-axis protrude from left and right side surfaces of the movable block **41**. The shock absorber **44** is disposed adjacent to a front surface of the push plate **411** with a piston rod **442** oriented toward the push plate **411**. The shock absorber **45** is disposed adjacent to a back surface of the push plate **411** with a piston rod **452** oriented toward the push plate **411**. A cylinder **441** of the shock absorber **44** and a cylinder **451** of the shock absorber **45** are fixed to the base **2**. A distal end of the piston rod **442** abuts against the operating surface **411***a* formed on the front surface of the push plate **411**, and a distal end of the piston rod **452** abuts against the

operating surface **411***b* formed on the back surface of the push plate **411**.
(33) Movement of the movable block **41** in the X-axis positive direction is damped mainly by the shock absorber **44**, and movement of the movable block **41** in the X-axis negative direction is damped mainly by the shock absorber **45**. By using the pair of shock absorbers **44** and **45** disposed in opposite directions, it is possible to more efficiently damp impact (vibration) of the movable block **41**.
(34) In the present embodiment, hydraulic shock absorbers are used as the shock absorbers **44** and **45**. However, pneumatic shock absorbers may be used instead of the hydraulic shock absorbers. Alternatively, the hydraulic shock absorber and the pneumatic shock absorber (or an elastic element such as an air spring or a coil spring) may be used by connecting them in series or in parallel. For example, as in a McPherson strut suspension of an automobile, a configuration may be adopted in which a shock absorber and a coil spring are coaxially disposed (i.e., the shock absorber is passed through a hollow portion of the coil spring) to connect the shock absorber and the coil spring in parallel.
(35) In the collision type test, an impact is applied to the dolly **20** and the test piece W by causing the dolly **20** that is inertially traveling to collide with the impact generating device **40**. At this time, the table **21** of the dolly **20** and the movable block **41** of the impact generating device **40** collide with each other via the pads **22** and **43**. Furthermore, the vibration (impact) of the movable block **41** generated by the collision is absorbed and attenuated by the shock absorbers **44** and **45**. Therefore, an impact waveform the impact generating device **40** applies to the test piece W changes depending on characteristics of the shock absorbers **44** and **45** (first shock absorbers) and the pads **22** and **43** (second shock absorbers). The characteristics of the shock absorbers **44** and **45** (the first shock absorbers) and the pads **22** and **43** (the second shock absorbers) are adjusted so that an impact having a desired waveform can be applied to the test piece W.
(36) The fall preventing structure of the present embodiment includes a first support **50** (first section) installed on the base **2** independently of the dolly **20**, and a second support **60** (second section) installed on the dolly **20**.
(37) The first support **50** includes a substantially gate-shaped support body **51**, and a pair of track units **52** (support body track units) that support the support body **51**. The support body **51** prevents the test piece W from falling over by coming into contact with the test piece W when the test piece W loses balance and tilts.
(38) In the non-collision type test, a position at which the test piece W falls over (i.e., a position in the X-axis direction at which a large impact acts on the test piece W) changes depending on the test condition. Therefore, the first support **50** of the present embodiment is configured such that the position of the support body **51** in the X-axis direction can be changed.
(39) As shown in FIG. **3**, the support body **51** has a beam **513** (a connecting portion) extending in the left-right direction, a pair of legs **512** extending downward from left and right end portions of the beam **513**, a pair of feet **51** extending forward from lower end portions of the respective legs **512**, and two pairs of ribs **514** coupling the legs **512** and the feet **511** to each other. It should be noted that the beam **513** and the pair of legs **512** of the present embodiment are integrally formed by cutting out from a single flat plate. As shown in an enlarged view surrounded by a two dot dashed line in FIG. **3**, a plurality of through holes **511***a* penetrating vertically are formed to the feet **511**. For example, the plurality of through holes **511***a* are formed in a row at equal intervals in the X-axis direction.
(40) A rectangular cutout portion **51***n* surrounded by the beam **513** and the pair of legs **512** is formed at a lower portion of the support body **51**. A width (Y-axis dimension) and the height (Z-axis dimension) of the cutout portion **51***n* are larger than those of the dolly **20** so that the dolly **20** can pass through the cutout portion **51***n*. That is, in a state in which the test piece W is not placed on the dolly **20**, it is possible to change an anteroposterior relationship (an arrangement order in the X-axis direction) between the dolly **20** and the support body **51**.

(41) As shown in FIG. **1**, the pair of track units **52** are installed at left and right end portions of the base **2**. As shown in an enlarged view in FIG. **3**, the track unit **52** includes a guide rail **521**, T-groove nuts **522**, and bolts **523**. The guide rail **521** is an elongated member (a so-called T-groove rail) to which a T-shaped groove extending in the X-axis direction is formed. A plurality of T-groove nuts **522** are fitted into the T-shaped groove of the guide rail **521**.
(42) The bolts **523** passed through the through holes **511***a* of the support body **51** are fitted into the T-groove nuts **522**, whereby the support body **51** is releasably fixed to the track unit **52**. That is, the track units **52** have a function as a fixing structure for fixing the support body **51** to the base **2**.
(43) When the bolts **523** are loosened, the fixing of the support body **51** to the track units **52** is released, and the support body **51** can move in the X-axis direction. At this time, since the bolts **523** are not removed from the T-groove nuts **522**, the feet **511** of the support body **51** remain coupled to the T-groove nuts **522** via the bolts **523**. Since the T-groove nuts **522** are fitted into the T-shaped groove, the T-groove nuts **522** (and the support body **51** coupled to the T-groove nuts **522**) can move only in the X-axis direction which is the extending direction of the T-shaped groove. In other words, when the fixing of the support body **51** to the track units **52** is released, the track units **52** function as a guide that guides the movement of the support body **51** in the X-axis direction.
(44) It should be noted that, in the present embodiment, the track units **52** are configured as a fixing and guide that functions as both a fixing structure that fixes the support body **51** to the base **2** and a guide that guides movement of the support body **51** in the X-axis direction (the traveling direction of the dolly **20**). However, the fixing structure and the guide may be provided separately.
(45) The second support **60** includes four post stands **62** installed near four corners on the table **21** of the dolly **20**, four posts **64** (columnar supports) detachably held by the respective post stands **62**, and a net **66** (a linear member) supported by the four posts **64**. The post stands **62** are cylindrical members extending vertically, and lower portions of the posts **64** are inserted into hollow portions of the post stands **62**.
(46) The net **66** is formed in a box shape (rectangular parallelepiped shape) with an open side facing downward, covers four sets of the post **64** and the post stand **62**, and is fixed to the post **64**, the post stand **62**, or the table **21** by a conventionally-known fixing member. The net **66** is formed for example of elastomer such as synthetic rubber and has stretchability. The posts **64** are for example installed so as to be lower than the test piece W. As a result, an upper portion of the test piece W comes into contact with the net **66** and is elastically and gently held by the net **66**, whereby the test piece W is prevented from falling over.
(47) By configuring the second support **60** from relatively thin members (or members formed by connecting thin members in a net-like fashion) such as the posts **64** and the net **66**, and arranging these thin members at intervals (e.g., at intervals larger than the thickness of each member), it is possible to reduce a weight of the second support **60**, and it is possible to reduce air resistance the second support **60** receives when the dolly **20** travels. This reduces the power required to drive the dolly **20** and makes it possible to reduce the capacity of the servomotor **342**, and thus it becomes possible to reduce power consumption and size of the impact test device **1**.
(48) When the height of the test piece W is low, the posts **64** may not be used, and the net **66** may be fixed to the post stand **62** or the table **21**.
(49) In place of the net-like member such as the net **66**, a string-like member such as a rubber string or a rope may be used as the linear member, and in place of the elastomer, a linear member formed of a material having normal elasticity (energy elasticity) such as polypropylene or steel may be used.
(50) A configuration may be adopted in which a linear member that does not have elasticity is used and a fixing member that elastically fixes the linear member to the post **64** or the dolly **20** is used.
(51) FIG. **6**(*a*) is a block diagram illustrating a schematic configuration of a control system **1***a* of the impact test device **1**. The control system **1***a* includes a controller **72** that controls operation of the entire device, a measurement unit **74** that performs various measurements, and an interface unit

76 that performs input and output with the outside.
(52) The servomotor **342** of each drive module **34** is connected to the controller **72** via a servo amplifier **342***a*. A rotary encoder RE is built in the servomotor **342**. Phase information of the shaft **342***b* of the servomotor **342** detected by the rotary encoder RE is input to the controller **72** via the servo amplifier **342***a*.
(53) The controller **72** and each servo amplifier **342***a* are communicably connected to each other with an optical fiber, and high-speed feedback control can be performed between the controller **72** and each servo amplifier **342***a*. Accordingly, it is possible to synchronously control the plurality of servomotors **342** with higher accuracy (specifically, with high resolution and high accuracy in the time axis).
(54) The interface unit **76** includes, for example, one or more of a user interface for performing input from and output to a user, network interfaces for connecting to various networks such as a LAN (Local Area Network), and various communication interfaces for connecting to external devices such as USB (Universal Serial Bus) and GPIB (General Purpose Interface Bus). The user interface includes one or more of various input/output devices such as, for example, various operation switches, indicators, various display devices such as an LCD (Liquid Crystal Display), various pointing devices such as a mouse and touch pad, a touch screen, a video camera, a printer, a scanner, a buzzer, a speaker, a microphone and a memory card reader/writer.
(55) The measurement unit **74** includes an acceleration sensor **742** and a linear encoder **744** attached to the dolly **20**. The measurement unit **74** generates measurement data by amplifying and digitally converting signals from the acceleration sensor **742** and the linear encoder **744**, and transmits the measurement data to the controller **72**. An acceleration sensor **742** to be attached to the test piece W can further be added to the measurement unit **74** to measure an impact acting on the test piece W during the test.
(56) The controller **72** synchronously controls driving of the servomotors **342** of the drive modules **34** on the basis of control conditions such as impact waveforms (e.g., acceleration waveforms) input through the interface unit **76** and measurement data input from the measurement unit **74**. In the present embodiment, two servomotors **342** are driven in the same phase (precisely, the servomotors **342** of the drive modules **34**FL and **34**BL on the left and the servomotors **342** of the drive modules **34**FR and **34**BR on the right are driven in opposite phases (in opposite rotating direction)).
(57) As described above, the impact test device **1** according to the present embodiment can be used to perform two types of tests, namely, the collision type test and the non-collision type test. Hereinafter, a content of and a procedure for each test will be described.
(58) [Collision Type Test]
(59) The collision type test is performed in a state in which the connection between the dolly **20** and the carriage **32** of the dolly driver **30** is released. In the collision type test, a force that causes the test piece W to fall over forward acts on the test piece W when the dolly **20** collides with the impact generating device **40**. Therefore, in a case where the first support **50** is used, for example, as shown in FIG. **1**, the support body **51** (more specifically, the beam **513**) of the first support **50** is installed in the vicinity of the rear end portion of the impact generating device **40** (i.e., the pads **43**). The position of the support body **51** of the first support **50** is appropriately adjusted in accordance with a size, shape, weight distribution, and the like of the test piece W.
(60) In the collision type test, first, for example, the carriage **32** is moved to a start position S set in the vicinity of a rear end of a travelable range (the movable range of the dolly **20** in the X-axis direction) by driving the drive module **34**. Then, the dolly **20** is moved, for example, manually, to a position where the dolly **20** contacts with the carriage **32**, and the test piece W is placed on the table **21** of the dolly **20**. When the second support **60** is used, the test piece W is held on the table **21** by the second support **60**.
(61) When loading of the test piece W on the dolly **20** is completed and a command to start a test is

issued through a user operation on, for example, a touch screen (the interface unit **76**) or the like, continuous measurement of an impact by the acceleration sensor **742** mounted on the dolly **20** starts. Detection results by the acceleration sensor **742** are accumulated and stored in a storage **721** connected to the controller **72**, is graphed, and is displayed on a display device (the interface unit **76**) as an impact waveform (e.g., an acceleration waveform).

(62) Then, the carriage **32** is gradually accelerated forward by the driving of the drive module **34**. At this time, the dolly **20** is pushed by the carriage **32** to travel at the same speed as the carriage **32**. When the carriage **32** reaches a preset collision speed, the carriage **32** is decelerated. As the carriage **32** decelerates, the dolly **20** moves away from the carriage **32**, inertially travels at the collision speed, and finally collides with the impact generating device **40**.

(63) In the collision type test, the impact generated by the collision between the dolly **20** and the impact generating device **40** acts on the test piece W placed on the dolly **20**. By the collision, a force to tilt the test piece W forward acts on the test piece W. Therefore, the test piece W tilts forward, but the test piece W is prevented from falling over since the test piece W contacts the support body **51** of the first support **50** disposed in the vicinity of the front end of the dolly **20**.

(64) When a predetermined time elapses after the collision, the measurement by the acceleration sensor **742** stops, the test piece W is taken out of the dolly **20**, thereby one collision type test ends.

(65) [Non-Collision Type Test]

(66) The non-collision type test is performed in a state in which the carriage **32** and the dolly **20** are integrally coupled by the coupling parts **321**.

(67) In the non-collision type test, a position where the test piece W is expected to fall over (an expected falling position) changes depending on test conditions (a shock waveform or the like). When using the first support **50**, the support body **51** of the first support **50** is installed in the vicinity of the expected falling position.

(68) Then, the dolly **20** coupled to the carriage **32** is moved to a start position by the drive of the drive module **34**. The start position of the non-collision type test may be set to a position different from the start position S of the collision type test (e.g., in the vicinity of a center of the travelable range of the dolly **20**). At the start position, the test piece W is placed on the table **21** of the dolly **20**. When the second support **60** is used, the test piece W is held on the table **21** by the second support **60**.

(69) When loading of the test piece W on the dolly **20** is completed and a command to start a test is issued through the user operation on the interface unit **76**, continuous measurement of an impact by the acceleration sensor **742** mounted on the dolly **20** starts.

(70) Then, driving of the servomotor **342** of each drive module **34** is controlled on the basis of waveform data of a preset impact waveform (e.g., an acceleration waveform). An impact generated by each drive module **34** is transmitted to the carriage **32** and the dolly **20** by the belt mechanism **35** and is applied to the test piece W placed on the dolly **20**.

(71) Since, for example, a force to tilt the test piece W forward acts on the test piece W due to the impact applied to the test piece W, the test piece W tilts forward. However, since the test piece W comes into contact with the support body **51** of the first support **50** disposed in the vicinity of the front end of the dolly **20**, the test piece W is prevented from falling over.

(72) When a predetermined time elapses after the application of the impact to the test piece W, the measurement by the acceleration sensor **742** stops, the test piece W is taken out of the dolly **20**, thereby one non-collision type test ends.

Second Embodiment

(73) Next, a second embodiment of the present disclosure will be described. An impact test device **2000** according to the second embodiment is a device in which the first support is made automatically movable. Hereinafter, mainly aspects of the second embodiment which are different from the first embodiment will be described, and redundant description of configurations common to the first embodiment will be omitted.

(74) FIG. **7** is a plan view of the impact test device **2000** according to the second embodiment of the present disclosure. A first support **2500** of the impact test device **2000** includes a support body **2510**, a track unit **2520** that supports the support body **2510** so as to be movable in the X-axis direction, and a support body driver that drives the support body **2510**. The support body driver includes four drive modules **2540**FL, **2540**FR, **2540**BL, and **2540**BR (hereinafter collectively referred to as "drive modules **2540**" as appropriate and each referred to as "drive module **2540**" without discrimination as appropriate) that generate power for driving the support body **2510**, and two sets of belt mechanisms **2550**R and **2550**L on the left and right (hereinafter collectively referred to as "belt mechanisms **2550**" as appropriate and each referred to as "belt mechanism **2550**" without discrimination as appropriate) that transmit the power generated by the drive modules **2540** to the support body **2510**. The support body **2510** is driven in the X-axis direction by the power generated by the drive modules **2540**.
(75) The four drive modules **2540** are disposed near four corners of the base **2**, respectively. The belt mechanism **2550**L on the left is driven by the pair of drive modules **2540**FL and **2540**BL on the left, and the belt mechanism **2550**R on the right is driven by the pair of drive modules **2540**FR and **2540**BR on the right. The configuration of the drive module **2540** is the same as the configuration of the drive module **34** of the first embodiment, and thus the description thereof will be omitted.
(76) The track unit **2520** includes a pair of roadbeds **2521** installed on the base **2** and extending in the X-axis direction, and a pair of linear guides **2522** provided on the roadbeds **2521**, respectively. Each linear guide **2522** includes a rail **2522***a* laid on an upper surface of the roadbed **2521**, and a plurality of (e.g., two) runner **2522***b* capable of traveling on the rail **2522***a*. The runners **2522***b* are attached to a lower surface of leg portions **2511** of the support body **2510**. The linear guide **2522** guides the movement of the support body **2510** in the X-axis direction.
(77) Each belt mechanism **2550** includes a pair of drive pulleys **2551**, a toothed belt **2552** wound around the pair of drive pulleys **2551**, and a belt clamp **2553** (winding intermediate fixing tool) that fixes the toothed belt **2552** to the support body **2510**. The drive pulleys **2551** are coupled to shafts **2544** of respective drive modules **2540**. The support body **2510** is provided with horizontally disposed flat plate-shaped clamp attachment parts **2515** protruding outward to the left and right from respective legs **2512**. The toothed belts **2552** are attached to the respective clamp attachment parts **2515** by belt clamps **2553**. The support body **2510** is driven forward and backward (in the X-axis direction) by the power transmitted by the drive module **2540**.
(78) As shown in FIG. **6**(*b*), a servomotor **2542** of each drive module **2540** is connected to the controller **72** via a servo amplifier **2542***a*. A rotary encoder RE is built in the servomotor **2542**. Phase information of a shaft of the servomotor **2542** detected by the rotary encoder RE is input to the controller **72** via the servo amplifier **2542***a*.
(79) Hereinafter, two examples related to drive control of the support body **2510** of the first support **2500** of the impact test device **2000** according to the second embodiment of the present disclosure will be described.
Example 1
(80) In Example 1, the movement and fixing of the support body **2510** of the first support **50** which are manually performed in the first embodiment are automated. Specifically, before starting the impact test (e.g., before placing the test piece W on the dolly **20**), the controller **72** synchronously controls the four drive modules **2540** (the servomotors **2542**) of the support body driver to move the support body **2510** to a predetermined position (support body installation position) set in accordance with the test condition.
(81) In the collision type test, the support body installation position is set, for example, at a position adjacent to the front of the dolly **20** disposed at a foremost end of a range in which the dolly **20** travels during the test (i.e., in the vicinity of the rear end portion of the impact generating device **40**). In the non-collision type test, the support body installation position is set, for example, at a

position adjacent to the front of the dolly **20** disposed at the foremost end of the range in which the dolly **20** travels during the test or at a position adjacent to the rear of the dolly **20** disposed at a rearmost end of the range in which the dolly **20** travels during the test. For example, when a sign of a maximum acceleration of an impact waveform (or an integral value of the acceleration) is positive (i.e., when a forward impact is to be applied), the support body installation position is set forward of the travel range of the dolly **20**, and when the sign is negative (i.e., when a rearward impact is to be applied), the support body installation position is set rearward of the travel range of the dolly **20**.

(82) In Example 1, during the test, the support body **2510** is not driven but is held at the support installation position. In other words, the support body driver functions as a fixing structure that releasably fixes the support body **2510** to the base **2** during the test.

Example 2

(83) In Example 2, during the test, the support body **2510** is not kept still, and the dolly **20** and the support body **2510** are driven while keeping a distance between the dolly **20** and the support body **2510** substantially constant. Specifically, the controller **72** synchronously controls the four drive modules **34** (specifically, the servomotors **342**) of the carriage drive section and the four drive modules **2540** (specifically, the servomotors **2542**) of the support body driver on the basis of an impact waveform.

(84) The driving of the four drive modules **2540** of the support body driver may be controlled on the basis of an impact waveform subjected to smoothing processing (e.g., simple moving average). By controlling the driving on the basis of the impact waveform subjected to smoothing processing, acceleration acting on the support body **2510** is alleviated, and power consumption of the servomotor **2542** is reduced.

(85) According to the illustrative embodiments of the present disclosure described above, since the first supports **50** and **2500** are separated from the dolly **20**, the increase in the weight of the dolly **20** and the air resistance during traveling due to the introduction of the fall preventing structure is reduced, and the increase in the power required to drive the dolly **20** is reduced. Therefore, it is possible to drive the dolly **20** by using a motor of a smaller capacity. In addition, since degradation in traveling characteristics (e.g., accuracy and stability of a traveling speed) and a change in impact characteristics (e.g., an impact waveform to be applied to the test piece W) of the dolly **20** due to the introduction of the fall preventing structure are reduced, degradation in the test accuracy can be suppressed. Furthermore, in the non-collision type test, a larger impact (acceleration) can be applied to the test piece W by suppressing the increase in the weight of the dolly (impact table) and the air resistance.

(86) Hereinabove, the illustrative embodiments according to aspects of the present disclosure have been described. The present disclosure can be practiced by employing conventional materials, methodology and equipment. Accordingly, the details of such materials, equipment and methodology are not set forth herein in detail. In the previous descriptions, numerous specific details are set forth, such as specific materials, structures, chemicals, processes, etc., in order to provide a thorough understanding of the present disclosure. However, it should be recognized that the present disclosure can be practiced without reapportioning to the details specifically set forth. In other instances, well known processing structures have not been described in detail, in order not to unnecessarily obscure the present disclosure.

(87) Only exemplary illustrative embodiments of the present disclosure and but a few examples of their versatility are shown and described in the present disclosure. It is to be understood that the present disclosure is capable of use in various other combinations and environments and is capable of changes or modifications within the scope of the inventive concept as expressed herein. For example, appropriate combinations of configurations of embodiments and the like explicitly illustrated in this specification and/or configurations that are obvious to a person with ordinary skills in the art from the description of this specification are also included in the embodiments of

this application. As further examples, according to aspects of the present disclosure, the following modifications are possible.

(88) In each of the embodiments described above, the dolly **20** and the carriage **32** are separated from each other so that both the collision type test and the non-collision type test can be performed. However, the dolly **20** and the carriage **32** may be integrated to form a device dedicated to the non-collision type test.

(89) In the above-described embodiments, the fall preventing structure includes both the first support **50** (the first support **2500**) and the second support **60**. However, the fall preventing structure may include only one of the first support and the second support.

(90) In each of the above-described embodiments, the toothed belt **352** is wound around the pair of drive pulleys **351**. However, one of the pulleys around which the toothed belt **352** is wound may be a driven pulley. In this case, one of the drive modules **34**FL (**34**FR) and **34**BL (**34**BR) is not necessary. Such configuration can also be applied to the belt mechanism **2550** of the support body driver.

(91) In each of the above-described embodiments, the carriage **32** is driven by the two toothed belts **352**. However, the carriage **32** may be driven by one or three or more toothed belts **352**. Such configuration can also be applied to the support body driver.

(92) FIG. **8** is a plan view schematically showing an exemplary configuration in which the carriage **32** is driven by four toothed belts **352**. As shown in FIG. **8**, lengths (number of teeth) of the respective toothed belts **352** may be made the same. By making the lengths of the toothed belts **352** the same, transmission characteristics of the plurality of toothed belts **352** become uniform, and thus the carriage **32** can be driven more stably. Such configuration can also be applied to the belt mechanism **2550** of the support body driver. Furthermore, by arranging the adjacent drive modules **34** in the front-rear direction (i.e., by shifting positions of the adjacent drive modules **34** with respect to each other in the font-rear direction), it is possible to make the impact test device compact.

(93) The pitch diameters (the number of teeth) of the respective drive pulleys **351** may be made the same. Reduction ratios of the belt mechanisms **343** built in the respective drive modules **34** may also be made the same. With such configuration, since ratios of driving amounts of the respective servomotors **342** (rotation angles of the respective shafts **342***b*) to moving amounts of the respective toothed belts **352** become the same, the servomotors **342** can be driven with the same driving amounts. These configurations can also be applied to the support body driver.

(94) FIG. **9** is a diagram showing a schematic configuration of a modified example of the drive module **34**. As shown in FIG. **9**, the drive module **34** may include a plurality of (for example, two) servomotor **342**. In the modified example shown in FIG. **9**, the shafts **342***b* of a pair of servomotors **342** are connected to both ends of a shaft **344** by couplings **346**, respectively. Thereby, one drive pulley **351** can be driven by the pair of servomotors **342**. A configuration in which a plurality of drive pulley **351** (belt mechanism **35**) are coupled to one shaft **344** may also be adopted. With this configuration, it is possible to change ratio of the servomotors **342** to the drive pulleys **351** coupled to the shaft **344**, and thus it is possible to effectively make use of performances of the servomotor **342** and the toothed belt **352**.

(95) In the above-described embodiment, a linear member having rubber elasticity is used. However, for example, a linear member formed of a highly elastic material such as polyamide-based resin or steel may be used.

Claims

1. An impact test device comprising: a controller; a dolly configured to travel with a test piece placed thereon; a dolly driver configured to drive the dolly; an impact generating device configured to collide with the dolly to generate an impact to be applied to the test piece; and a fall preventing

structure configured to prevent the test piece from falling over, wherein the fall preventing structure includes a first section independent of the dolly, the first section is movable in a traveling direction of the dolly, the first section comprises: a support body configured to support the test piece when the test piece tilts to prevent the test piece from falling over, a guide configured to guide movement of the support body in the traveling direction, a support body driver configured to drive the support body in the traveling direction, the impact test device is configured to perform: a collision type test in which the impact is applied to the test piece by causing the dolly on which the test piece is placed to collide with the impact generating device, and a non-collision type test in which an impact waveform is applied to the test piece by transmitting the impact waveform, which is generated by driving the dolly driver, to the dolly, and the controller is configured to: control the support body driver and the dolly driver, move the support body to a predetermined position set in accordance with a test condition, control the dolly driver to cause the dolly to collide with the impact generating device at a predetermined speed in the collision type test, and control the dolly driver so that the dolly is driven based on the impact waveform in the non-collision type test.

2. The impact test device according to claim 1, comprising a base on which the first section is installed, wherein the first section includes a fixing structure configured to releasably fix the support body to the base.

3. The impact test device according to claim 2, wherein the first section includes a fixing and guide serving as both the guide and the fixing structure, the fixing and guide comprising: a T-shaped groove fixed to the base and extending in the traveling direction; a T-groove nut fitted into the T-shaped groove; and a bolt that is inserted in a through hole formed to the support body and fitted into the T-groove nut, thereby fixing the support body to the T-shaped groove.

4. The impact test device according to claim 1, wherein the support body driver comprises: a drive module configured to generate power for driving the support body; and a belt mechanism configured to transmit the power generated by the drive module to the support body.

5. The impact test device according to claim 1, wherein the dolly driver includes: a carriage releasably coupled to the dolly; and a dolly track configured to travellably support the dolly and the carriage, in the non-collision type test, the carriage is coupled to the dolly, and in the collision type test, connection between the carriage and the dolly is released.

6. The impact test device according to claim 1, wherein the impact generating device comprises: a movable block; a first plastic programmer attached to a surface of the movable block facing the dolly; a block track configured to support the movable block so as to be movable in the traveling direction; and a shock absorber configured to absorb vibration of the movable block, the block track includes a linear guide, the linear guide comprises: a rail; and a runner attached to the movable block and capable of traveling on the rail via rolling elements, and the dolly comprises a second plastic programmer attached to a surface facing the movable block.

7. The impact test device according to claim 1, wherein the fall preventing structure includes a second section installed on the dolly, and the second section comprises: a plurality of columnar support erected on the dolly; and a linear member stretched over the plurality of columnar support.

8. The impact test device according to claim 7, wherein the linear member includes one of a string-like member having a string shape and a net-like member having a net shape.